

Analytics Vidya Course Search Engine Report

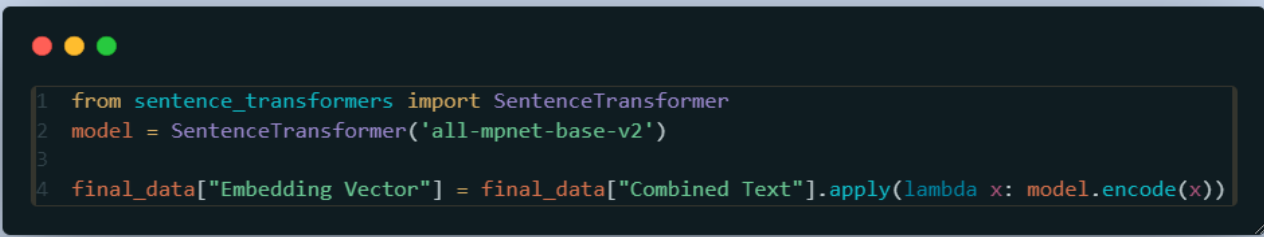
Project Files: <https://github.com/Satwik-uppada/Analytics-Vidya/tree/main>

1. **model.ipynb:** This file handles data preparation and model development.
2. **app.py:** This file implements the Streamlit web application for the search engine.
3. **analytics_vidhya_courses.csv:** This file contains Course name, image url, course url, lessons, price.
4. **course_details.csv:** This file contains course name, description, curriculum.

Data Preparation and Model Development (model.ipynb)

The model.ipynb file performs the following steps:

- **Data Loading:** Loads course data from two CSV files:
 - analytics_vidhya_courses.csv: Contains basic course information like title, lessons, price, image URL, and course link.
 - course_details.csv: Contains detailed course information like description and curriculum.
- **Data Cleaning:**
 - Filters the data to include only free courses.
 - Handles missing values in the Lessons column by filling them with an empty string.
- **Data Merging:** Merges the two dataframes based on the Course Name column.
- **Curriculum Flattening:** Converts the dictionary-like structure of the Curriculum column into a readable string.
- **Combined Text Generation:** Creates a new column "Combined Text" by concatenating the Course Name, Description, and Curriculum columns.
- **Embedding Generation:** Uses the **"all-mpnet-base-v2" SentenceTransformer model** to generate embedding vectors for each course based on the Combined Text.



```
1 from sentence_transformers import SentenceTransformer
2 model = SentenceTransformer('all-mpnet-base-v2')
3
4 final_data["Embedding Vector"] = final_data["Combined Text"].apply(lambda x: model.encode(x))
```

- **Data Saving:** Saves the processed data to a CSV file final_Data.csv.

- **Elasticsearch Indexing:**

Elastic Search Cloud: [search-tool - Elastic](#)

Elastic Search URL: <https://my-elasticsearch-project-d00981.es.us-east-1.aws.elastic.cloud:443>

- Creates an Elasticsearch index named search-tool with appropriate mappings for each column.

The screenshot shows the Elasticsearch Index Management interface for an index named 'search-tool'. The breadcrumb navigation indicates the path: My Elasticsearch project > Data > Index Management > search-tool. The left sidebar contains navigation links for Home, Data, Index Management (selected), Connectors, Build, Dev Tools, Playground, Analyze, and Discover. The main content area displays the index details:

- Elasticsearch URL:** `https://my-elasticsearch-project-d00981.es.us-east-1.aws.elastic.cloud:443`
- API Key:** [Redacted]
- Document count:** 116
- Index Size:** 2.42mb
- AI Search:** 1 Field

Sparse Vector	0 Fields
Dense Vector	1 Field
Semantic Text	0 Fields

- Indexes the course data into the Elasticsearch index.

Streamlit Web Application (app.py)

The app.py file implements the Streamlit web application for the search engine. It provides the following functionalities:

Huggingface Space: https://huggingface.co/spaces/Satwikuu/Smart_search_tool

- **User Interface:**

- Displays a title "Course Search Engine" and a sidebar with a logo and search tips.
- Provides a text input field for users to enter their search query.
- Includes a sidebar checkbox to enable debug mode.

- **Search Functionality:**

- Initializes an Elasticsearch client.
- Corrects spelling mistakes in the user query using the `correct_spelling` function.
- Performs a search using the `search_courses` function, which:
 - ✓ Splits the query into terms for highlighting.
 - ✓ Lemmatizes and removes stop words from the query.
 - ✓ Constructs a complex Elasticsearch query using `bool` and `should` clauses to prioritize matches in different fields.
 - ✓ Deduplicates results based on Course Link while keeping the highest scoring version.
 - ✓ Highlights search terms in the course name, description, and curriculum using HTML.

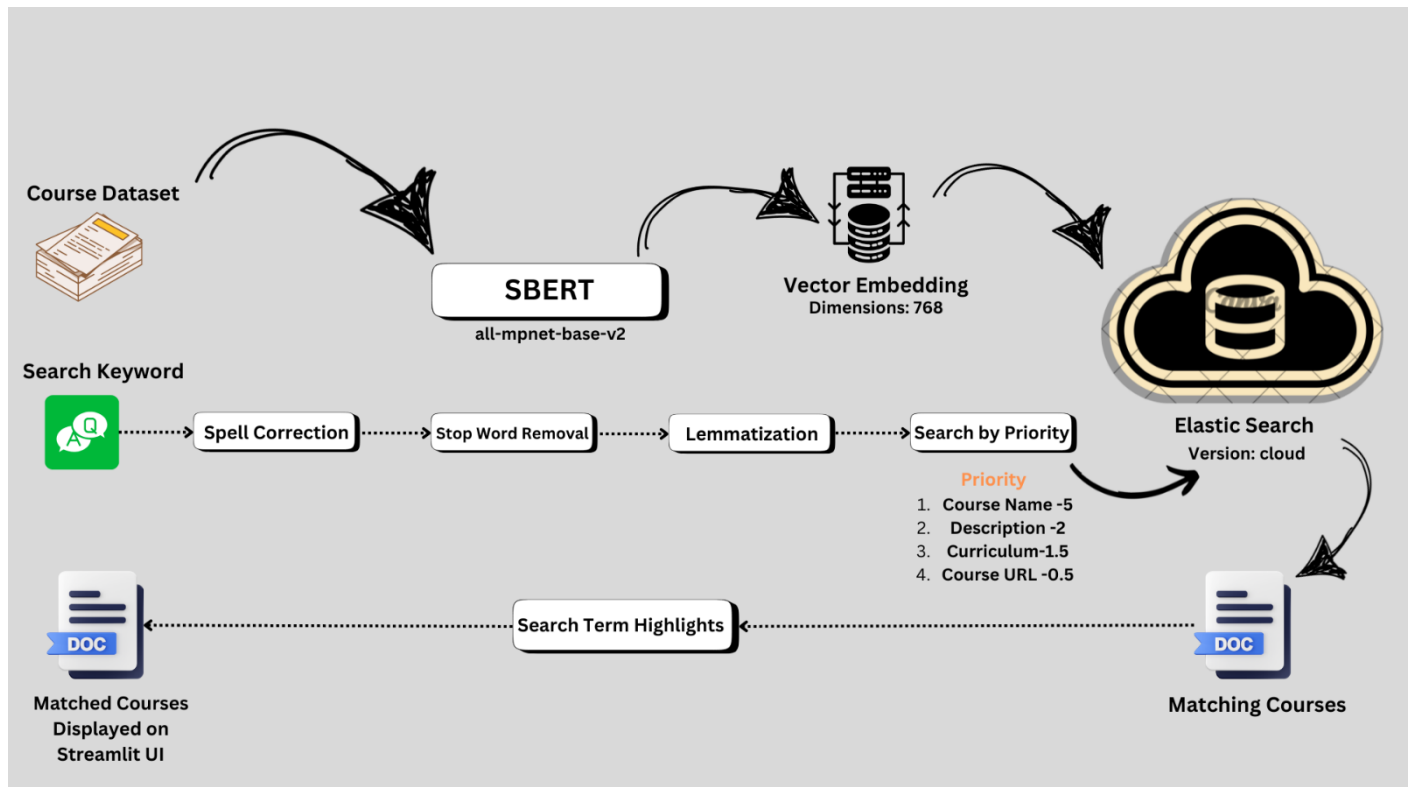
- **Result Display:**

- Displays the search results in a card-like format.
- Shows the course name, description, and curriculum (if available) with highlighted search terms.
- Includes a "View Course Details" button that links to the course page.
- Provides debug information (if debug mode is enabled) including search score, course URL, and total vs. unique matches.

Overall Analysis

The Analytics Vidya Course Search Engine is a well-designed application that leverages the power of Elasticsearch and SentenceTransformers for efficient and accurate course search. The code is well-structured and includes features like spelling correction, query term highlighting, and result deduplication to enhance the user experience. The Streamlit web application provides a user-friendly interface for searching and browsing courses.

Process Flow Chart:



Advanced Semantic Search of Courses using SBERT and Elastic Search

Web App UI:

The screenshot shows the 'Smart_search_tool' web application. On the left sidebar, the 'Enable Debug Mode' checkbox is checked and highlighted with a red box, with an arrow pointing to the text 'Debug mode on'. Below this, 'Search Tips' and 'Debug Information' are listed. The main content area is titled 'Course Search Engine' with a magnifying glass icon. It features a search input field containing 'mahine learnin', which is highlighted with a red box and an arrow pointing to the text 'Wrong spelled word'. Below the input field, a green bar displays 'Searching for machine learning', also highlighted with a red box and an arrow pointing to 'After Spell Correction'. The search results show 'Machine Learning Summer Training', with 'Machine Learning' highlighted in a red box and an arrow pointing to 'Search term highlighter'. The description below the title says 'This is the second step of the Machine Learning Summer Training, want to know more click here.', with 'Machine Learning' highlighted in a red box. A dropdown menu labeled 'Curriculum' is open, showing 'Debug Info:' with a red box around it and an arrow pointing to 'Debug Mode details'. The debug info includes 'Search Score: 96.05' and a URL: 'https://courses.analyticsvidhya.com/courses/machine-learning-summer-training'. At the bottom, the text 'Machine Learning Certification Course for Beginners' is displayed.

Reference:

1. SBERT: https://sbert.net/docs/sentence_transformer/pretrained_models.html
2. Elastic cloud: <https://www.elastic.co/guide/en/cloud/current/ec-getting-started.html>
3. <https://medium.com/@abidsaudagar/build-semantic-search-with-elasticsearch-and-bert-vector-embeddings-from-scratch-41707cb4e2fd>
4. <https://medium.com/@abidsaudagar/building-advanced-semantic-search-with-elasticsearch-and-gpt-3-5-turbo-28647e155fda>